



Determination of antiacetylcholinestrasic and antioxidant properties from octocorals of Andaman and Nicobar Islands

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Abstract: This research was conducted with the intention of determining whether or not octocorals have antiacetylcholinestrasic and antioxidant capabilities. Utilising the α , α -Diphenyl- β -PicrylHydrazyl (DPPH) radical-scavenging and *in vitro* acetylcholinesterase test, this objective was successfully achieved. As positive controls, we used ascorbic acid and galanthamine. Crude extracts were produced from *Isis* sp.; *Rumphella* sp.; *Melithaea* sp. 1.; *Melithaea* sp. 2.; and *Carijoa riisei* (Duchassaing & Michelotti, 1860). These extracts were acquired by the use of methanol extraction. It showed that gorgonian corals, which belong to the order Malacalcyonacea and the family Melithaeidae, specifically *Melithaea* sp.1, possess a significant amount of antiacetylcholinesterase activity that is dose-dependent. On the other hand, alcyonian corals, which belong to the family Carijoidae, specifically *C. riisei*, have been reported to have a high level of antioxidant activity. *Isis* sp. has a far greater antioxidant capacity than the other crude octocoral extracts investigated in this study. Additionally, this study provides the first evidence that members of the *Melithaea* genus have antiacetylcholinesterase capabilities.

Résumé : Détermination des capacités antiacétylcholinestérases et antioxydantes des octocaralliaires des Iles Andaman et Nicobar. Cette recherche a été menée dans le but de déterminer si les octocaralliaires ont ou non des capacités antiacétylcholinestérases et antioxydantes. En utilisant le test de piégeage des radicaux α , α -Diphényl- β -PicrylHydrazyl (DPPH) et le test *in vitro* d'acétylcholinestérase, cet objectif a été atteint avec succès. Comme contrôles positifs, nous avons utilisé l'acide ascorbique et la galanthamine. Des extraits bruts ont été produits à partir d'*Isis* sp.; *Rumphella* sp.; *Melithaea* sp. 1.; *Melithaea* sp. 2.; et *Carijoa riisei* (Duchassaing & Michelotti, 1860). Ces extraits ont été obtenus par extraction au méthanol. Il a été démontré que les coraux gorgones, qui appartiennent à l'ordre des Malacalcyonacea et à la famille des Melithaeidae, en particulier *Melithaea* sp.1, possèdent une quantité significative d'activité antiacétylcholinestérase qui est dose dépendante.

D'autre part, les coraux alcyonnaires, qui appartiennent à la famille Carijoidae, et plus précisément *C. riisei*, ont été signalés comme ayant un niveau élevé d'activité antioxydante. *Isis* sp. a une capacité antioxydante bien supérieure à celle des autres extraits bruts d'octocoralliaires étudiés ici. De plus, cette étude fournit la première preuve que les membres du genre *Melithaea* ont des capacités antiacétylcholinestérases.

Keywords: Acetylcholinesterase inhibition • Antioxidative potential • Bioactive compound • Gorgonians • Melithaeidae

Introduction

The exploration of marine bioactive compounds has gained significant momentum since the mid-1960s, resulting in the discovery of nearly 20,000 unique compounds. Among these, octocorals, which include soft corals, gorgonians, sea pens, and blue corals, have emerged as a particularly rich source of bioactive substances. The orders Alcyonacea and Scleractinia within the class Anthozoa are noted for their contributions to the pool of promising marine compounds, showcasing a diverse array of secondary metabolites with potential therapeutic applications. These organisms are not only visually striking but also rich in nutritionally important substances, which may play a crucial role in their ecological interactions and survival strategies. The incidence of predation in the majority of these organisms is low due to the toxic compounds they produce to deter predators (Hooper & Coleman, 1995). Extensive biosynthetic studies have been conducted on octocoral metabolites, revealing a plethora of compounds with significant pharmaceutical potential, including antifoulants and various bioactive agents (Bhakuni & Rawat, 2005). Notably, octocorals are known to produce a variety of secondary metabolites such as diterpenes, sesquiterpenes, furanoditerpenes, terpenoids, capnellenes and steroids (Konig & Wright, 1998; Chang et al., 2008; Grote et al., 2008). According to Yabe et al. (2000), stolonidiol, a bioactive marine diterpene isolated from an unidentified genus *Clavularia* (F. Clavulariidae), showed potent choline acetyltransferase (ChAT) inducible activity in primary cultured basal forebrain cells and clonal septal SN49 cells.

In recent years, bioactive compounds have attracted considerable attention in pharmacology due to their potential inhibitory activity against cholinesterase enzymes, particularly acetylcholinesterase (AChE) and butyrylcholinesterase (BChE). These enzymes play a crucial role in the hydrolysis of acetylcholine, a neurotransmitter essential for cognitive functions. Inhibition of cholinesterase activity is a well-established therapeutic strategy for managing neurodegenerative diseases, especially Alzheimer's disease (AD), where cholinergic transmission is impaired (Noetzi & Eap, 2013). The relevance of bioactive compounds as cholinesterase inhibitors lies in their ability to enhance

synaptic cholinergic activity, thereby alleviating symptoms associated with cognitive decline (Biscussi et al., 2019). Research has demonstrated that various plant extracts and their bioactive constituents exhibit significant cholinesterase inhibitory activity. For instance, extracts from *Morinda morindoides* have shown promising inhibition of AChE and BChE, indicating their potential as natural therapeutic agents for managing conditions like AD (Sinan et al., 2020). Similarly, triterpenes isolated from mushrooms such as *Fuscoporia torulosa* have been reported to possess potent cholinesterase inhibitory properties, further supporting the notion that natural products can serve as effective cholinesterase inhibitors (Deveci et al., 2019). Additionally, phenolic compounds derived from vegetables like cabbage and cucumber have been identified as effective cholinesterase inhibitors, highlighting the diverse sources of bioactive compounds with therapeutic potential (Obboh et al., 2017). The pharmacological applications of these inhibitors extend beyond AD. They may also be beneficial in treating other conditions characterized by cholinergic dysfunction, such as myasthenia gravis and glaucoma (Biscussi et al., 2019). Moreover, the development of novel cholinesterase inhibitors, including those derived from natural sources, is crucial for addressing the limitations of existing synthetic drugs, which may have adverse side effects (Pinho et al., 2013). For instance, the selective inhibition of cholinesterases can lead to improved cognitive function without the hepatotoxic risks associated with some synthetic compounds (Scheiner et al., 2020).

Gorgonian corals have been proven to be sources of interesting natural terpenoid derivatives (Berrue et al., 2009; Blunt et al., 2012). According to Chung et al. (2009), two sesquiterpenoid phenols were isolated from the Formosan gorgonian coral *Echinomuricea* sp. (family Plexauridae). Studies performed to find natural antioxidants indicated that many gorgonian species have this potential. In addition, studies have already been shown that antioxidant preparations from gorgonian coral extracts can prevent cancer, as well as cardiovascular, neurodegenerative, inflammation and aging related diseases (Matulja et al., 2021).

Since, the Class Octocorallia; order Malacalcyonacea and Scleralcyonacea are the ones which have contributed with the highest number of promising bioactive marine compounds, Zoological

Survey of India, has recently turned its attention on five octocorals from three different families and species *Isis* sp. (Isididae), *Rumphella* sp., *Melithaea* sp. 1, *Melithaea* sp.2 (Melithaeidae) and *Carijoa riisei* (Duchassaing & Michelotti, 1860) (Clavulariidae) for evaluating their antioxidant (scavenging of free radicals and superoxide anion) and antiacetylcholinesterase (acetylcholinesterase inhibitor) activities. As IC₅₀ is

Materials and Methods

A graphical representation of the overall strategy and methodology used in the present study is shown in figure 1. The antiacetylcholinesterase and antioxidant tests were therefore performed using a series of chemicals: Ellman's reagent -5,5'-dithio-bis-2-nitrobenzoic acid (DTNB), Acetylcholine esterase

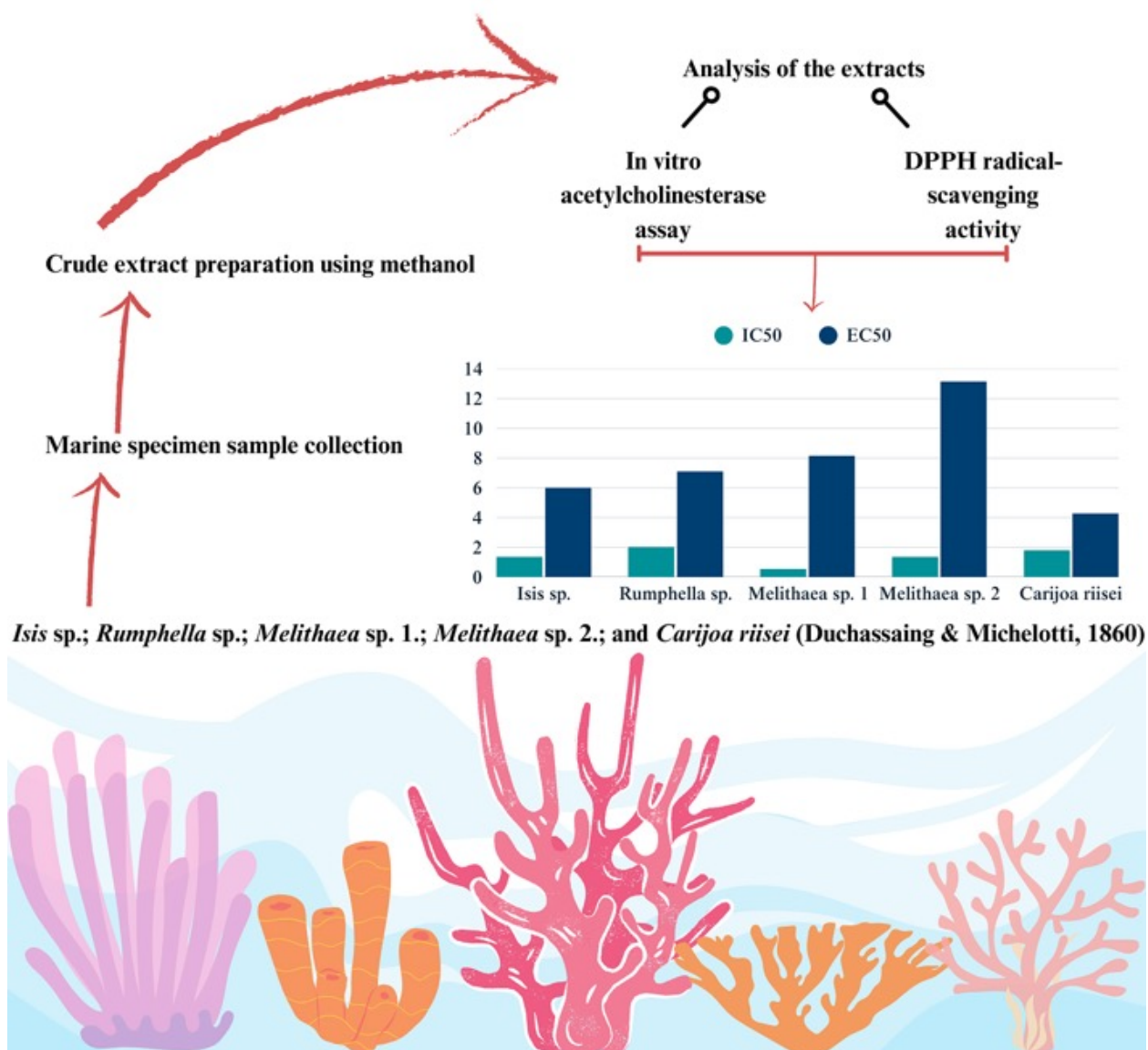


Figure 1. Graphical representation of the methodology of the study.

a quantitative measure which indicates how much of a substance is needed to inhibit a given biological process or component of a process by half, the present study aims to display the values of IC₅₀ (half maximal inhibitory concentration) for these octocorals (Aykul & Martinez-Hackert, 2016; Kelutur et al., 2021).

(AChE) (Electrophorous electricus AChE Type VI-S, EC 3.1.1.7, Sigma), ATCI (Acetylthiocholine iodide), Galanthamine hydrobromide, 1, 1-diphenyle-2-picrylhydrazyl (DPPH), L-ascorbic acid, Dimethyl sulfoxide (DMSO).

Biological materials

Specimens of the octocoral species were collected by along the coast of Andaman and Nicobar Islands and stored in a freezer until extraction.

The voucher specimens were deposited in the National Zoological Collection, Zoological Survey of India (ANRC), Port Blair. These organisms were identified as follows (Fig. 2):

- Sample 1: *Isis* sp. (O: Malacalcyonacea; F: Isididae)
- Sample 2: *Rumphella* sp. (O: Malacalcyonacea; F: Isididae)
- Sample 3: *Melithaea* sp. 1 (O: Malacalcyonacea; F: Melithaeidae)

- Sample 4: *Melithaea* sp. 2 (O: Malacalcyonacea; F: Melithaeidae) and
- Sample 5: *Carijoa riisei* (Duchassaing & Michelotti, 1860) (O: Malacalcyonacea; F: Carijoidae)

Preparation of crude extracts

About 20 g of tissue were extracted from the five different species using the solvent methanol (80% methanol, 1:20, w/v) at room temperature $27 \pm 2^\circ\text{C}$, at atmospheric pressure by shaking the sample at 100 rpm for 72 hours. Each extract was filtered and concentrated by using a rotary evaporator. The concentrated extracts were freeze-dried and stored in a refrigerator till used.

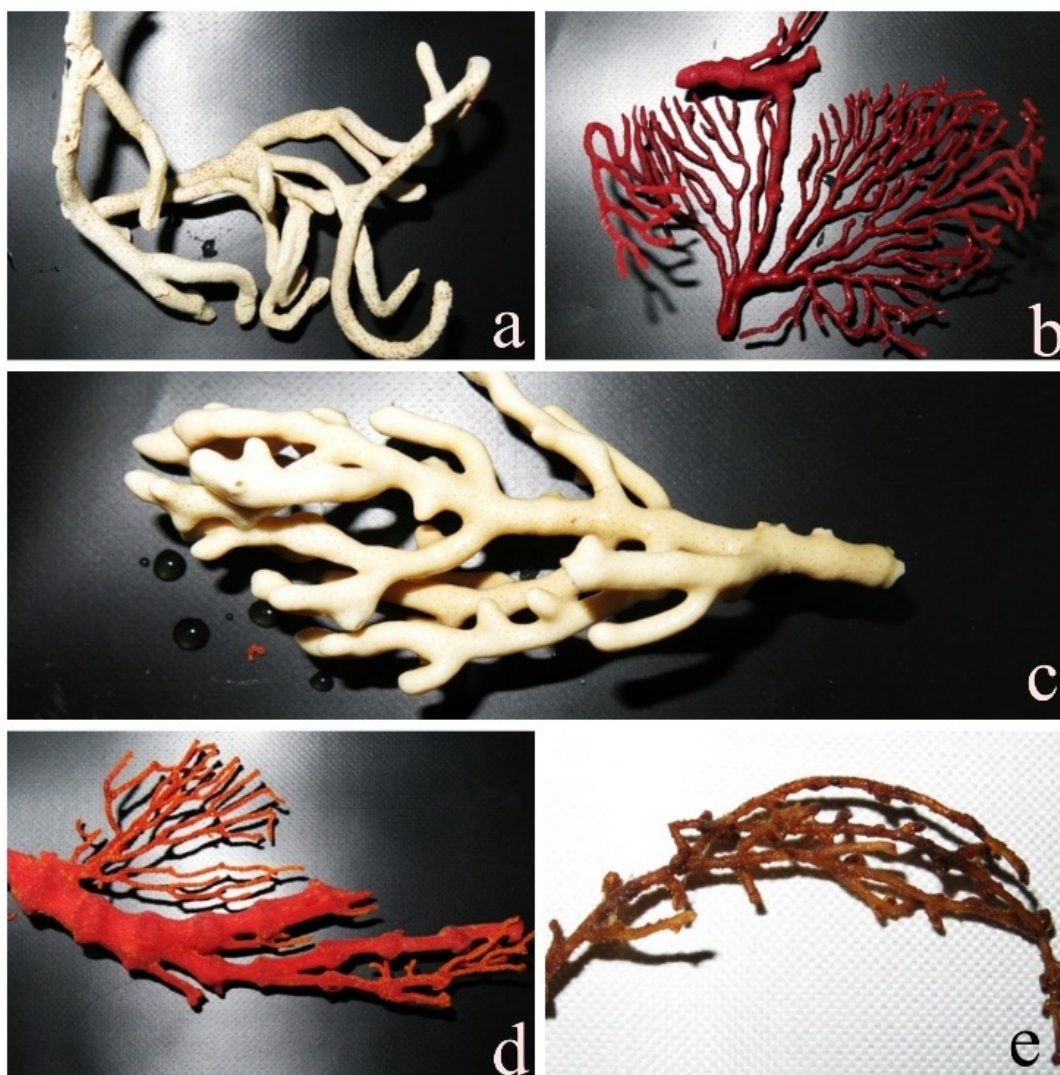


Figure 2. Pictures showing octocoral samples used for the study. **A.** *Rumphella* sp. **B.** *Melithaea* sp. 1. **C.** *Isis* sp. **D.** *Melithaea* sp. 2. **E.** *Carijoa riisei* (Duchassaing & Michelotti, 1860).

In vitro acetylcholinesterase (AChE) assay

AChE inhibitory activity was measured by the spectrophotometric method previously reported by Ellman et al. (1961), modified by Rhee et al. (1991) using a 96-well microtiter plate reader and adapted according to the present experimental conditions. In order to calculate the activity of the obtained acetylcholinesterase the following procedure was employed: each well of the 96-well microtiter plate, 1 μ L of test compound and 10 μ L of 0.01 U.mL⁻¹ of acetylcholine esterase were incubated at 4°C for 20 min before the addition of 10 μ L of 1.5 mM ATCI (Acetylthiocholine iodide) in phosphate buffer (pH 7.5), 60 μ L of 3 mM 5,5'-dithio-bis-2-nitrobenzoic acid (DTNB) in phosphate buffer (pH 7.5), 60 μ L of 0.1 M phosphate buffer (pH 7.5) and absorbance was measured at 405 nm every 30 s for 4 min $\times$ 8 times on a TRIAD multimode reader (Dynex Technologies, Inc., Chantilly, VA). Galanthamine hydrobromide (10 μ M) solution in DMSO solvent was used as positive control and each assay was performed in triplicate. The enzyme hydrolyses the substrate acetylthiocholine resulting in the product thiocholine which reacts with Ellman's DTNB reagent *in situ* and gives the yellow-coloured chromophore of 5-thio-2-nitrobenzoic acid (TNB) which can be detected at 405 nm. The absorption intensity of TNB adduct (405 nm) is proportional to the formation of thiocholine, therefore to the AChE activity. Percentage of inhibition was calculated by comparing the optical density for each sample with respect to the blank (10% DMSO in buffer). The percentage (%) inhibition of AChE activity was calculated using the formula: $[(A_{\text{control}} - A_{\text{sample}})/A_{\text{control}}] \times 100$.

DPPH radical-scavenging activity

The DPPH radical-scavenging activity [11] was tested for the extracts and L-ascorbic acid (standard) at different concentrations in methanol. About 3.5 ml of DPPH (0.06 mM in methanol) was added to each of these solutions (0.1 ml). After further mixing, the decrease in absorbance was recorded at 515 nm on a UV-visible double beam spectrophotometer (Lambda 25, Perkin-Elmer, Shelton, CT) after 30 min incubation, against methanol without DPPH as the blank reference. Absorbances were converted to the DPPH radical-scavenging rate according to the equation:

DPPH radical scavenging rate (%): $[(A_{\text{control}} - A_{\text{sample}})/A_{\text{control}}] \times 100$. The anti-free radical activity was defined as the relative concentration of antioxidant required to lower the initial DPPH concentration by 50% [IC₅₀ (mol.L⁻¹ compound per unit DPPH concentration)]. About 3.5 ml of 0.06 mM DPPH solution along with 0.1 mL of methanol was used as control, and 3.5 mL

of methanol along with 0.1 mL of sample solution was added as blank. All experiments were carried out at room temperature.

Results and Discussion

Antiacetylcholinesterase (acetylcholinesterase inhibitor) activity

The AChE inhibition of samples was analyzed in terms of IC₅₀ values in mg.mL⁻¹ obtained in comparison to that of galanthamine. The concentration of the crude extract that inhibited 50% of AChE activity (IC₅₀) was calculated using Microsoft Excel software.

Among the specimens tested, *Melithaea* sp. 1 (*M. ochracea*) exhibited the most potent activity with an IC₅₀ of 0.54 ± 0.06 mg.mL⁻¹, demonstrating a stronger inhibitory effect compared to other octocoral extracts (Fig. 3). *Isis* sp. and *Melithaea* sp. 2 showed moderate activity with IC₅₀ values of 1.37 ± 0.27 mg.mL⁻¹ and 1.35 ± 0.15 mg.mL⁻¹, respectively. *Rumphella* sp. and *C. riisei* displayed higher IC₅₀ values of 2.03 ± 0.2 mg.mL⁻¹ and 1.81 ± 0.7 mg.mL⁻¹, indicating lower activity. The standard compound, galantamine, exhibited significantly greater activity with an IC₅₀ of 0.000920 ± 0.0009 mg.mL⁻¹, confirming its effectiveness as a reference inhibitor. This highlights the potential of *Melithaea* sp. 1 as a promising source of anti-acetylcholinesterase agents, although further purification and characterization of active compounds are needed to enhance its efficacy (Simoni et al., 2012). However, it is the first time that this property has been reported for *Melithaea* sp. 1. Also, the species *C. riisei* showed potent choline acetyltransferase (ChAT) inhibitory activity. The activities were dose-dependent. Lower concentrations of *Melithaea* sp. 1 were needed to inhibit at least 50% the AChE activity, whereas the inhibitory effect of other crude extracts needed more elevated concentrations. The poorest activity was obtained with the extracts of *Rumphella* sp.

The dose-dependent nature of the inhibitory effects observed (Fig. 4) in this study aligns with existing literature, which consistently demonstrates that varying concentrations of extracts significantly influence AChE activity. For instance, (Koca et al., 2022) reported that the potency of AChE inhibitors is often contingent upon the concentration of the extract utilized, with lower concentrations yielding more pronounced inhibitory effects. This finding is corroborated by other studies that have shown a similar trend, where the efficacy of natural extracts in inhibiting AChE is directly related to their concentration, emphasizing the importance of optimizing dosages for therapeutic applications (Çavdar et al., 2019). The particularly low IC₅₀ value of *Melithaea* sp. 1, which required only 0.54 ± 0.06 mg.L⁻¹

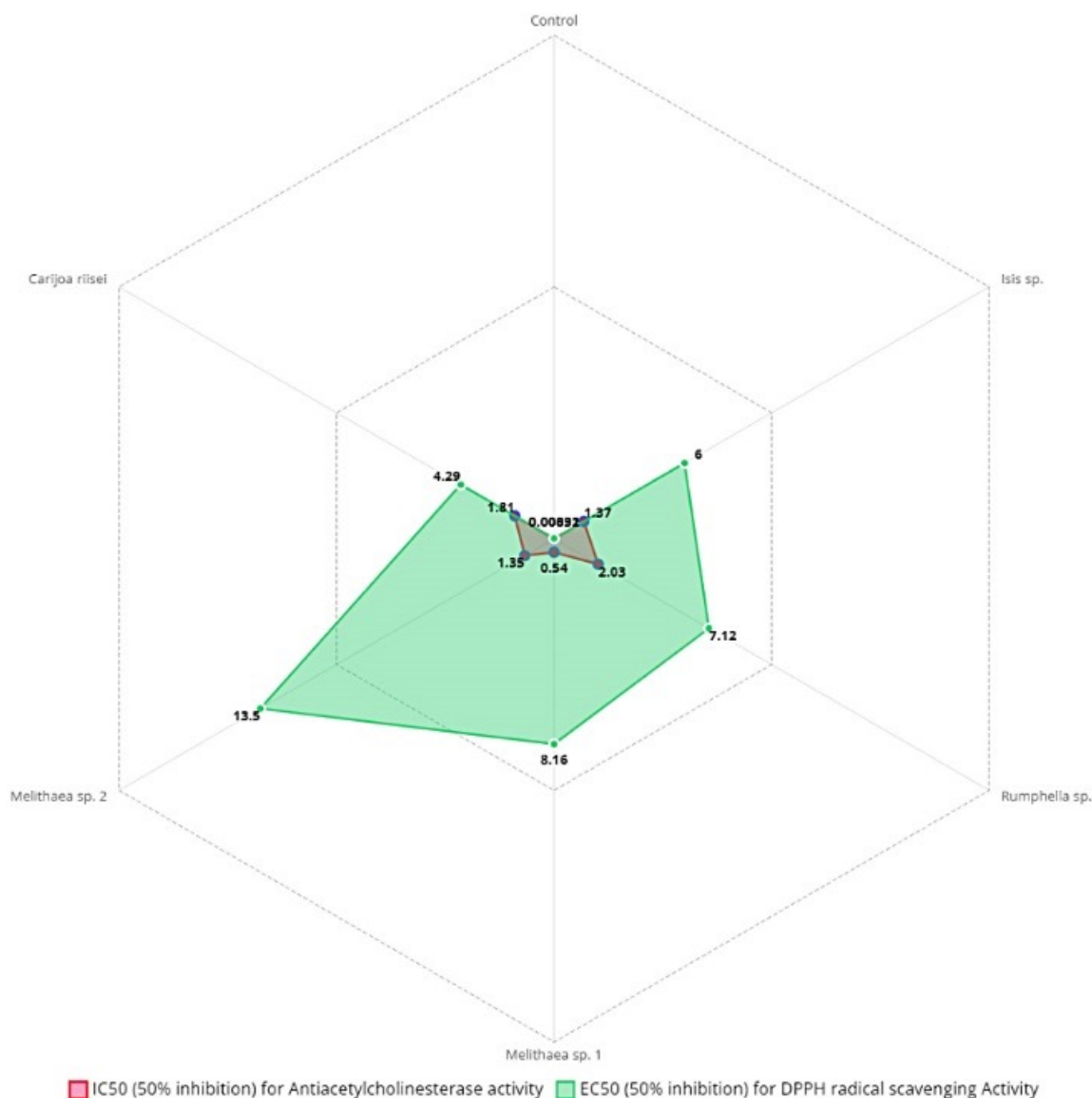


Figure 3. Graph showing IC₅₀ (50% inhibition) of crude extract of octocorals and Galanthamine for Antiacetylcholinesterase activity; and showing EC₅₀ (50% inhibition) of crude extract of octocorals and ascorbic acid for DPPH radical scavenging activity.

to achieve 50% inhibition of AChE activity, suggests that this species may harbour highly effective bioactive compounds. This is consistent with findings from Gülçin et al. (2016), who noted that marine organisms, especially corals, are rich sources of novel AChE inhibitors that could be pivotal in developing treatments for Alzheimer's disease and other neurodegenerative conditions (Gülçin et al., 2016). The potential for *Melithaea* sp. 1 to yield bioactive compounds that can be isolated and characterized for therapeutic use is promising, as the discovery of such compounds could contribute significantly to the pharmacological arsenal against cognitive decline. While numerous plant extracts exhibit greater potency, this study highlights

the AChE-inhibitory properties of soft corals through their crude extracts. These findings pave the way for the isolation of specific active components, potentially contributing to therapeutic advancements, which may be more potent compared to other bioactive compounds. Moreover, the observation that *C. riisei* exhibited potent choline acetyltransferase (ChAT) inhibitory activity adds another dimension to the therapeutic potential of these coral extracts. ChAT is essential for synthesizing acetylcholine, and its inhibition could enhance the neuroprotective profile of extracts from this species, as suggested by (Lee & Jun, 2019). The dual inhibition of AChE and ChAT could provide a synergistic effect, potentially leading

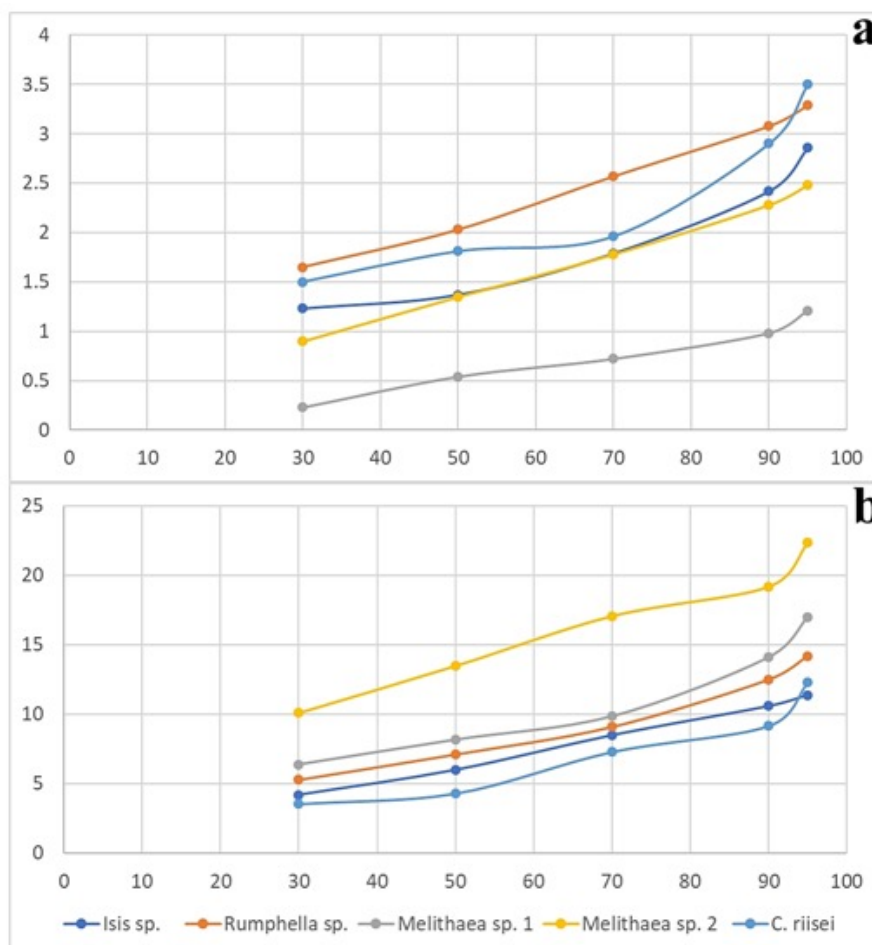


Figure 4. Dose dependant curve of crude extract of octocorals. **A.** Antiacetylcholinesterase activity. **B.** DPPH radical scavenging activity.

to improved cholinergic function and cognitive outcomes, which is particularly relevant in the context of Alzheimer's disease where cholinergic deficits are a hallmark (Klug et al., 2018). The significance of AChE inhibitors in treating neurodegenerative diseases is well-documented, with various studies emphasizing the role of these compounds in enhancing acetylcholine levels in the synaptic cleft, thereby improving neurotransmission and cognitive function (Çavdar et al., 2019). For example, Xu et al. (2023) highlighted that selective AChE inhibition could lead to increased acetylcholine concentrations, which is crucial for alleviating symptoms associated with Alzheimer's disease. Furthermore, the exploration of natural sources for AChE inhibitors, such as the extracts from *Melithaea* sp. 1 and *C. riisei*, aligns with the growing interest in developing safer and more effective therapeutic agents from marine biodiversity (Mahmood et al., 2021).

Anti-oxidant activity

The antioxidant potential of methanolic extracts from different octocorals was assessed using the DPPH free radical scavenging assay. The extract of *C. riisei* demonstrated the strongest antioxidant activity among the tested species, with an EC₅₀ of 4.29 ± 0.57 mg.mL⁻¹. This activity, while significant, was lower compared to the standard antioxidant ascorbic acid, which showed an EC₅₀ of 0.00831 ± 0.00033 mg.mL⁻¹, aligning with Wang et al. (2021). Other species exhibited varying levels of antioxidant activity, with *Isis* sp. showing an EC₅₀ of 6.00 ± 0.7 mg.mL⁻¹, followed by *Rumphella* sp. (7.12 ± 0.72 mg.mL⁻¹), *Melithaea* sp. 1 (8.16 ± 0.94 mg.mL⁻¹), and *Melithaea* sp. 2 (13.5 ± 1.27 mg.mL⁻¹). The dose-dependent increase in DPPH radical scavenging activity further supports the potential of these extracts as sources of natural antioxidants. Additionally, *Isis* sp. demonstrated notable acetylcholinesterase inhibitory activity (EC₅₀ = 6.00 ± 0.07 mg.mL⁻¹), aligning with previous studies on *I. hippuris*, where metabolites like sesquiterpenes, steroids, and isishippuric acid

B exhibited potent bioactivities, including cytotoxicity (Sheu et al., 2000 & 2004; Bhatnagar et al., 2012). The dose-dependent increase in DPPH radical scavenging activity observed in these extracts (Fig. 4) supports their potential as sources of natural antioxidants, which is consistent with findings in other studies that have reported similar antioxidant activities in marine organisms (Thảo et al., 2015). The presence of bioactive compounds such as terpenes and phenolics in these extracts may contribute to their free radical scavenging abilities, as these compounds are known for their antioxidant properties (Saragosti et al., 2010). This dual functionality of antioxidant and AChE inhibitory activities in octocorals suggests a promising avenue for further investigation into their bioactive compounds, particularly in the context of developing therapeutic agents for neurodegenerative diseases such as Alzheimer's (Karasová et al., 2008). The antioxidant mechanisms in corals, including the role of compounds like dimethylsulfoniopropionate (DMSP), have been documented as critical for coping with oxidative stress (Lellolter et al., 2018). DMSP and its breakdown products have been shown to scavenge reactive oxygen species (ROS), thereby contributing to the overall antioxidant capacity of coral holobionts (Deschaseaux et al., 2014). This highlights the ecological and physiological importance of these compounds in maintaining coral health, particularly under stress conditions such as temperature fluctuations and salinity changes (Maire et al., 2023).

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